

Beyond the ANOVA Procedure

STAT 200 – Introductory Statistics

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Today's Objective

After today's lecture, you should:

- Understand the theory behind testing:
 - equality means of more than two populations.
 - whether a categorical variable helps explain a numeric variable.
 - independence between a numeric and a categorical variable.
- determine if ANOVA or the Kruskal-Wallis test should be used
- determine which level is different using Tukey's HSD or the Kruskal multiple comparisons test
- explain the p-value and how to test hypotheses

Review: Procedure Requirements

Always, we have to make assumptions in determining the distribution of the test statistic. In many cases, you saw the assumptions. In some, they were hidden.

Test	Assumption
Z-test	Normality
t -test	Normality
Variance Test	Normality
Wilcoxon Test	Symmetry
Mann-Whitney Test	none
Binomial Test	none
Proportions Test	Normality
Chi-Square Goodness-of-Fit Test	Normality

ANOVA Requirements

For the analysis of variance procedure, there are two assumptions that we needed to make:

- Normality in each group.
- Equal variances across the groups.

These assumptions can be tested using the following tests:

- Shapiro-Wilk test (`shapiroTest`)
- Fligner-Killeen test (`fligner.test`)

The null hypothesis for each test is:

- H_0 : the data are from a NORMAL distribution.
- H_0 : The variances are equal. ($\sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$)

Thus, small p -values indicate the requirement is not met by the data.

Example #1: Rice Yields

Example

Does rice variety influence average yield amongst these four varieties?

First, because we are testing for independence between a numeric variable (yield) and a categorical variable (variety), we would like to use the analysis of variance procedure.

The hypotheses are:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_A : At least one of the means differs from the others.

Example #1: Rice Yields

We would like to use the analysis of variance procedure, because it is the most powerful test for this type of hypothesis. However, this test has two requirements:

- Normality in each group.
- Equal variances across the groups.

The following code loads the data and tests these two assumptions:

```
source("http://rfs.kvasaheim.com/stat200.R")
rice = read.csv("http://rfs.kvasaheim.com/data/rice.csv")
attach(rice)

shapiroTest(yield ~ variety)
fligner.test(yield ~ variety)
```

Example #1: Rice Yields

Here is the resulting output:

```
> shapiroTest(yield ~ variety)
```

```
$Title
```

```
[1] "Shapiro-Wilk Test"
```

```
$adjustment
```

```
[1] "Bonferroni (4)"
```

```
$results
```

	Level	p.value
1	A	0.3148640
2	B	1.0000000
3	C	0.5065811
4	D	0.8381207

Example #1: Rice Yields

Here is the resulting output:

```
> fligner.test(yield ~ variety)
```

```
Fligner-Killeen test of homogeneity of variances
```

```
data: yield by variety
```

```
Fligner-Killeen:med chi-squared = 1.0026, df = 3, p-value = 0.8006
```


Example #1: Rice Yields

Incomplete Conclusion:

We are asked to determine if the rice yield and the rice variety are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a NORMAL distribution in each group; and the variances are the same across the groups. Neither assumption is violated. The minimum p -value from the Shapiro-Wilk test is 0.0787, which is greater than $\alpha = 0.05$. The p -value from the Fligner-Killeen test, 0.8006, is also greater than our $\alpha = 0.05$. Because neither assumption is violated, we can use the analysis of variance procedure.

The small p -value of 0.00503 of the ANOVA procedure indicates that the two variables, yield and variety, are dependent. Not all varieties have the same average yield.

Beyond ANOVA

Note that the conclusion was *at least one mean differed from the others*. That is hardly helpful.

What we really want to know is “*which* variety is different?”

ANOVA cannot answer that question.

Tukey's HSD Test

However, there is a procedure that *can* answer it: Tukey's Honestly Significant Difference Test.

Its assumptions are the same as for ANOVA. So, if you use ANOVA, then you can use Tukey's HSD test.

Tukey's HSD Test

The code to perform Tukey's HSD test on our model is
“`TukeyHSD(ricemod)`”.

```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = yield ~ variety)
```

```
$variety
```

	diff	lwr	upr	p adj
B-A	-56.25	-191.592699	79.0927	0.6185496
C-A	-46.00	-181.342699	89.3427	0.7473470
D-A	132.00	-3.342699	267.3427	0.0567296
C-B	10.25	-125.092699	145.5927	0.9957690
D-B	188.25	52.907301	323.5927	0.0066015
D-C	178.00	42.657301	313.3427	0.0097522

Tukey's HSD Test

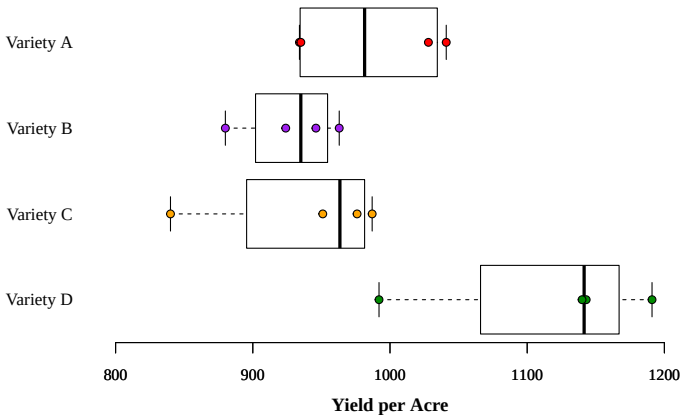
	diff	lwr	upr	p adj
B-A	-56.25	-191.592699	79.0927	0.6185496
C-A	-46.00	-181.342699	89.3427	0.7473470
D-A	132.00	-3.342699	267.3427	0.0567296
C-B	10.25	-125.092699	145.5927	0.9957690
D-B	188.25	52.907301	323.5927	0.0066015
D-C	178.00	42.657301	313.3427	0.0097522

Interpreting this abbreviated output:

- Find the *p*-values and interpret them for the null hypothesis “the difference in averages between these two levels is 0.”
- At our usual level of significance, we were only able to detect differences in average yield between Varieties B and D and between Varieties C and D.
- Variety D has a higher average yield than Variety B by between 53 and 324.
- Variety D has a higher average yield than Variety C by between 43 and 313.
- We did not detect a difference between average yield between any other comparisons.

Tukey's HSD Test

These conclusions are hardly surprising in light of the data:



Kruskal-Wallis Test

If any of the assumptions necessary for ANOVA are violated, you should *not* use ANOVA.

Like the t -test, there is a non-parametric alternative to ANOVA. It is called the **Kruskal-Wallis test**.

- It has the same requirements as the Mann-Whitney test.
- It is a part of the [agricolae](#) package, which you have to install.
- Its output is simpler than that of ANOVA.
- It has its own multiple comparisons test, the Kruskal (multiple comparisons) test.
- The Kruskal test output is also simpler than that of Tukey's HSD.

Example #2: Citizen's Initiative

Example

The citizen's initiative allows the people of the state to force a vote on a given issue. Do the different political cultures use the initiative at the same rates?

The hypotheses are:

$$H_0 : \mu_m = \mu_i = \mu_t$$

H_A : At least one of the means differs from the others.

Example #2: Citizen's Initiative

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a NORMAL distribution in each group.
- The data have the same variance across the groups.

To check this using R, we run the following code:

```
shapiroTest(inituse ~ domPolCulture)
flinger.test(inituse ~ domPolCulture)
```

Example #2: Citizen's Initiative

Here are the results:

```
$adjustment  
[1] "Bonferroni (4)"
```

```
$results  
  Level    p.value  
1 Individualistic 1.357651e-03  
2      Moralistic 2.048182e-03  
3 Traditionalistic 9.940607e-06
```

and:

Fligner-Killeen test of homogeneity of variances

```
data:  inituse by domPolCulture  
Fligner-Killeen:med chi-squared = 10.007, df = 2, p-value = 0.0067
```

Example #2: Citizen's Initiative

Because *at least one* requirement was not met, we should not use ANOVA. We will use the Kruskal-Wallis test:

```
kruskal.test(inituse, domPolCulture)
```

The resulting output is:

```
Kruskal-Wallis rank sum test
```

```
data:  inituse and domPolCulture
```

```
Kruskal Wallis chi-squared = 6.3238, df = 2, p-value = 0.04235
```

Because the p -value is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #2: Citizen's Initiative

To determine *which is different*, we use Kruskal's multiple comparisons test:

```
print(kruskal(inituse, domPolCulture))
```

The partial output is:

\$groups		inituse	groups
Moralistic	32.32353	a	
Individualistic	24.91176	ab	
Traditionalistic	20.76471	b	

Treatments with the same letter are not significantly different.

From this, we know we detected a difference in average initiative use between the moralistic states and the traditionalistic states. No other differences were detected.

Example #2: Citizen's Initiative

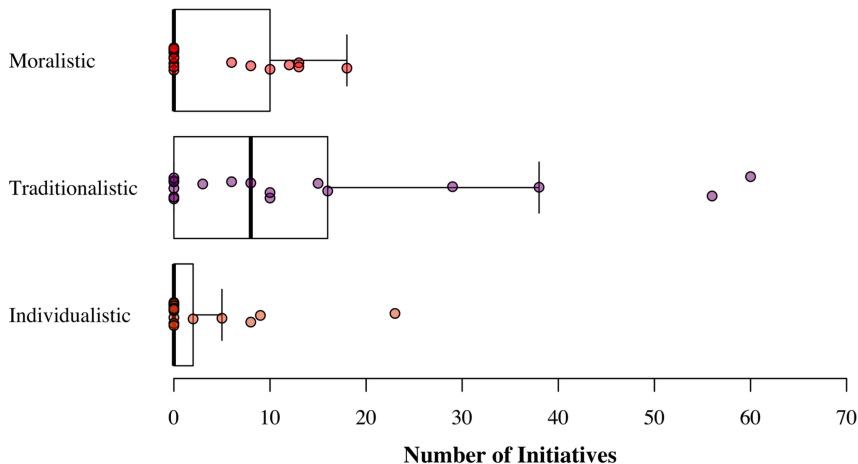
Conclusion:

We are asked to determine if the initiative use and the political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a NORMAL distribution in each group; and the variances are the same across the groups. Both assumptions are violated. The maximum p -value from the Shapiro-Wilk test is 0.00204, which is much less than $\alpha = 0.05$. The p -value from the Fligner-Killeen test, 0.0067, is also less than our $\alpha = 0.05$. Because both assumptions are violated, we should not use the analysis of variance procedure. We must use the Kruskal-Wallis test.

The small p -value of 0.0424 of the Kruskal-Wallis test indicates that the two variables, initiative use and political culture are dependent. Not all political cultures use the initiative process equally. According to the Kruskal multiple comparisons test, we can conclude that states with a moralistic political culture tend to use the initiative more frequently than states with a traditionalistic culture. No other comparisons were significant.

Example #2: Citizen's Initiative

Again, these conclusions make sense when seeing the graphic:



Example #3: Violent Crime

Example

Does the 2000 violent crime rate significantly vary across the four census regions?

The hypotheses are:

$$H_0 : \mu_N = \mu_S = \mu_M = \mu_W$$

H_A : At least one of the means differs from the others.

Example #3: Violent Crime

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a NORMAL distribution in each group.
- The data have the same variance across the groups.

To check this using [R](#), we run the following code:

```
shapiroTest(vcrime00 ~ census4)
flinger.test(vcrime00 ~ census4)
```


Example #3: Violent Crime

Here are the results:

```
$adjustment  
[1] "Bonferroni (4)"
```

```
$results  
      Level  p.value  
1  Midwest 1.0000000  
2 Northeast 1.0000000  
3    South 0.2251474  
4     West 1.0000000
```

and:

Fligner-Killeen test of homogeneity of variances

```
data: vcrime00 by census4
```

```
Fligner-Killeen:med chi-squared = 6.0913, df = 3, p-value = 0.1073
```

Example #3: Violent Crime

Because no requirement was violated, we can use ANOVA. Here is the code:

```
cmod1 = aov(vcrime00 ~ census4)
summary(cmod1)
```

and the results:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
census4	3	689569	229856	4.855	0.00505 **
Residuals	47	2225268	47346		

Because the p -value of 0.00505 is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Again, which one?

Example #3: Violent Crime

To determine which is different, we use the Tukey HSD test (`TukeyHSD(cmod1)`).

The partial output is:

```
$census4
```

	diff	lwr	upr	p adj
Northeast-Midwest	-55.21111	-310.76041	200.33819	0.9389452
South-Midwest	196.87647	-21.62826	415.38120	0.0910331
West-Midwest	-74.66154	-306.65974	157.33667	0.8266966
South-Northeast	252.08758	13.18663	490.98853	0.0350137
West-Northeast	-19.45043	-270.75207	231.85122	0.9968573
West-South	-271.53801	-485.05941	-58.01661	0.0075817

From this, we now we detected a difference in average violent crime rate between the South and Northwest states and the West and South states, where the South is significantly higher than the Northeast and the South is significantly higher than the West.

Example #3: Violent Crime

Conclusion:

We are asked to determine if the violent crime rate and the census region are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a NORMAL distribution in each group; and the variances are the same across the groups. Neither assumption is violated. The minimum p -value from the Shapiro-Wilk test is 0.2251, which is greater than $\alpha = 0.05$. The p -value from the Fligner-Killeen test, 0.1073, is also greater than our $\alpha = 0.05$. Because neither assumption is violated, we can – and should – use the analysis of variance procedure.

The small p -value of 0.00505 of the ANOVA test indicates that the two variables, violent crime rate and census region are dependent. Not all census regions have the same average violent crime rate. According to Tukey's HSD test, we can conclude that southern states have a significantly higher average violent crime rate than the Northeastern states and the Western states. No other comparisons were significant.

Example #4: Education

Example

Does the 2000 violent crime rate significantly vary across the four census regions?

The hypotheses are:

H_0 : All the means are the same.

H_A : At least one of the means differs from the others.

Example #4: Education

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a NORMAL distribution in each group.
- The data have the same variance across the groups.

To check this using R, we run the following code:

```
shapiroTest(waea00 ~ domPolCulture)
flinger.test(waea00 ~ domPolCulture)
```

Example #4: Education

Here are the results:

```
$adjustment  
[1] "Bonferroni (3)"
```

```
$results  
  Level    p.value  
1 Individualistic 1.0000000  
2      Moralistic 0.6071567  
3 Traditionalistic 1.0000000
```

and:

Fligner-Killeen test of homogeneity of variances

```
data: waea00 by domPolCulture
```

```
Fligner-Killeen:med chi-squared = 2.8698, df = 2, p-value = 0.2381
```

Example #4: Education

Because no requirement was violated, we can use ANOVA. Here is the code:

```
waeamod = aov(waea00 ~ domPolCulture)
summary(waeamod)
```

and the results:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
domPolCulture	2	276.0	138.02	17.19	2.35e-06 ***
Residuals	48	385.4	8.03		

Because the p -value is much less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #4: Education

To determine which is different, we use the Tukey HSD test (`TukeyHSD(waeamod)`).

The partial output is:

```
$domPolCulture
```

	diff	lwr	upr	p adj
Moral-Indiv	1.725882	-0.6247889	4.076554	0.1885858
Trad-Indiv	-3.840588	-6.1912595	-1.489917	0.0007316
Trad-Moral	-5.566471	-7.9171418	-3.215799	0.0000019

From this, we now we detected a difference in average WAEA rate between the traditionalist states and both of the other two types. In both cases, the traditionalistic states tended to have lower WAEA. No other comparisons were statistically significant.

Example #4: Education

Conclusion:

We are asked to determine if the violent crime rate and the census region are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a NORMAL distribution in each group; and the variances are the same across the groups. Neither assumption is violated. The minimum p -value from the Shapiro-Wilk test is 0.6072, which is greater than $\alpha = 0.05$. The p -value from the Fligner-Killeen test, 0.2381, is also greater than our $\alpha = 0.05$. Because neither assumption is violated, we can – and should – use the analysis of variance procedure.

The small p -value of 2.35×10^{-6} of the ANOVA test indicates that the two variables, WAEA and dominant political culture are dependent. According to Tukey's HSD test, we can conclude that traditionalistic states have a significantly lower average WAEA than the moralistic and individualistic states. No other comparisons were significant.

Example #5: Average Wealth

Example

Does the 2000 GSP per capita significantly vary across the three political cultures?

The hypotheses are:

$$H_0 : \mu_I = \mu_M = \mu_T$$

H_A : At least one of the means differs from the others.

Example #5: Average Wealth

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a NORMAL distribution in each group.
- The data have the same variance across the groups.

To check this using R, we run the following code:

```
shapiroTest(gspcap00 ~ domPolCulture)
flinger.test(gspcap00 ~ domPolCulture)
```

Example #5: Average Wealth

Here are the results:

```
$adjustment  
[1] "Bonferroni (3)"
```

```
$results  
      Level      p.value  
1 Individualistic 7.609321e-03  
2      Moralistic 7.771441e-01  
3 Traditionalistic 3.239349e-05
```

and:

Fligner-Killeen test of homogeneity of variances

data: gspcap00 by domPolCulture

Fligner-Killeen:med chi-squared = 0.7711, df = 2, p-value = 0.6801

Example #5: Average Wealth

Because at least one requirement was violated, we should not use ANOVA. We need to use the Kruskal-Wallis test. Here is the code:

```
kruskal.test(gspcap00 ~ domPolCulture)
```

and the results:

```
Kruskal-Wallis rank sum test
```

```
data:  gspcap00 by domPolCulture
```

```
Kruskal-Wallis chi-squared=13.752 , df=2, p-value=0.0010
```

Because the p -value of 0.0010 is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #5: Average Wealth

o determine *which is different*, we use Kruskal's multiple comparisons test:

```
print(kruskal(gspcap00, domPolCulture))
```

The partial output is:

```
$groups
          gspcap00 groups
Individualistic 36.35294      a
Moralistic      23.82353      b
Traditionalistic 17.82353      b
```

Treatments with the same letter are not significantly different.

From this, we now we detected a difference in average GSP per capita between the individualistic states and both of the other two types. In both cases, the individualistic states tended to have higher GSP per capita. No other comparisons were statistically significant.

Example #5: Average Wealth

Conclusion:

We are asked to determine if the average GSP per capita and the dominant political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a NORMAL distribution in each group; and the variances are the same across the groups. The Normality assumption is violated. The minimum p -value from the Shapiro-Wilk test is 0.000033, which is much less than $\alpha = 0.05$. Because an assumption is violated, we need to use the Kruskal-Wallis test.

The small p -value of 0.0010 of the Kruskal-Wallis test indicates that the two variables, GSP per capita and dominant political culture are dependent. According to the Kruskal multiple comparison test, we can conclude that individualist states have a significantly higher GSP per capita than the moralistic and traditionalistic. No other comparisons were significant.

Example #6: Legislature Professionalism

Example

Does the professional level of the state's legislature significantly vary across the three political cultures?

The hypotheses are:

$$H_0 : \mu_I = \mu_M = \mu_T$$

H_A : At least one of the means differs from the others.

Example #6: Legislature Professionalism

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a NORMAL distribution in each group.
- The data have the same variance across the groups.

To check this using R, we run the following code:

```
shapiroTest(profleg ~ domPolCulture)
flinger.test(profleg ~ domPolCulture)
```

Example #6: Legislature Professionalism

Here are the results:

```
$adjustment  
[1] "Bonferroni (3)"
```

```
$results
```

	Level	p.value
1	Individualistic	0.033975358
2	Moralistic	0.002017848
3	Traditionalistic	1.000000000

and:

Fligner-Killeen test of homogeneity of variances

data: profleg by domPolCulture

Fligner-Killeen:med chi-squared = 5.1794, df = 2, p-value = 0.0750

Example #6: Legislature Professionalism

Because at least one requirement was violated, we should not use ANOVA. We need to use the Kruskal-Wallis test. Here is the code:

```
kruskal.test(profleg ~ domPolCulture)
```

and the results:

```
Kruskal-Wallis rank sum test
```

```
data:  profleg by domPolCulture
```

```
Kruskal-Wallis chi-squared=6.5777 , df=2, p-value=0.0373
```

Because the p -value of 0.0373 is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #6: Legislature Professionalism

o determine *which is different*, we use Kruskal's multiple comparisons test:

```
print(kruskal(profleg, domPolCulture))
```

The partial output is:

\$groups	profleg	groups
Individualistic	32.82353	a
Moralistic	22.35294	b
Traditionalistic	21.06250	b

Treatments with the same letter are not significantly different.

From this, we now we detected a difference in average level of legislative professionalism between the individualistic states and both of the other two types. In both cases, the individualistic states tended to have higher levels of legislative professionalism. No other comparisons were statistically significant.

Example #6: Legislature Professionalism

Conclusion:

We are asked to determine if the level of legislative professionalism and the dominant political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a NORMAL distribution in each group; and the variances are the same across the groups. The Normality assumption is violated. The minimum p -value from the Shapiro-Wilk test is 0.0020, which is much less than $\alpha = 0.05$. Because an assumption is violated, we need to use the Kruskal-Wallis test.

The small p -value of 0.0373 of the Kruskal-Wallis test indicates that the two variables, legislative professionalism and dominant political culture, are dependent. According to Kruskal's multiple comparisons test, we can conclude that individualistic states tend to have a significantly higher level of legislative professionalism than the moralistic and traditionalistic states. No other comparisons were significant.